Handling Big Data with Open Source Frameworks

Big Data is a term used to define large scale data sets that are too complex to be manipulated with basic DBMS. Handling Big Data requires sophisticated hardware and software technologies. In this article, we shall discuss some of the most advanced open source technologies that can be used to store, clean, organise, analyse and retrieve Big Data.

Big Data is everywhere. You don't need me to tell you this unless you have been living under a rock for the past decade. Just one look at the annual global data size (as shown in Figure 1) demonstrates the fact perfectly. You may have an idea that Big Data is 'big'. But in order to transform that idea into a realisation, allow me to enumerate some interesting facts.

1) From the year 2010 to 2020, user interactions with data have gone up by a whopping 5000 per cent.
2) There are roughly 3.5 billion searches made on Google daily.
3) Users exchange over 65 billion messages a day on WhatsApp.
4) You, I and almost everyone we know has each generated roughly 1.7 megabytes of data per second in the year 2020.
5) Roughly 45 per cent of the businesses in the world have to allocate resources to handle Big Data on the cloud.
6) Big Data analytics is one the highest paying jobs in the tech industry. This is especially true in the case of big banks, big pharma and the Silicon Valley.
7) The Internet generates roughly 2.5 quintillion bytes of data per day.
8) According to the popular *Executives Survey* conducted by New Vantage, over 97 per cent of the organisations in the world are investing money and other resources in Big Data management.

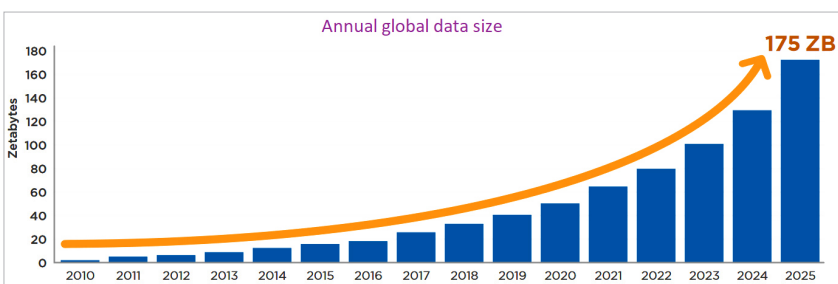

Figure 1: Annual global data size (Courtesy: Google Images)

So, yes, Big Data is extremely big. Here's another fun fact — most of the Big Data we generate never gets